Project Interim Report

On

File Compression Simulator (Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming with Java

SESSION: 2025-26

Submitted By

Suraj Kumar Sahu

ERP : RU-25-11498

Bachelor of Computer Application &

Data Science in association with Google

Under the Guidance of

Mr. Harsh Multani

Java EXPERT

SCHOOL OF INFORMATION TECHNOLOGY

RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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Abstract:

1. **What is your project:**

This project is a Global Currency Converter made using Java concepts. It converts foreign currencies like USD, EUR, and GBP into Indian Rupees (INR). All exchange rates are stored as fixed values, and the conversion is done through a simple method.
The Global Currency Converter is a simple and useful application developed using Java and Object-Oriented Programming (OOP) concepts. The main purpose of this project is to convert different international currencies such as USD, EUR, and GBP into Indian Rupees (INR) using predefined exchange rates. This project mainly focuses on the use of
static variables and static methods, which allows the program to perform operations without creating any object.

2. **Why is it needed:**

This project is needed because currency conversion is very useful in daily life, especially for people who travel, do online shopping, or are involved in international business. It helps users quickly find the value of one currency in another without doing manual calculations, saving time and reducing errors.

3. **What it does:**

The system works by taking an amount and a currency type as input from the user. Then checks the currency and applies the correct exchange rate to convert it into INR. Finally, it displays the converted result on the screen. It also handles invalid inputs by showing an error message. Overall, this project is simple, useful, and efficient.

Introduction:

In today's world, currency conversion is very important for people involved in travel, international business, online shopping, and financial transactions. Manually converting currency can be time-consuming and may lead to errors. This project provides a fast, accurate, and user-friendly solution to this problem. The system works by taking input from the user, such as the amount and the type of currency. It then uses stored exchange rates to calculate the equivalent value in INR and displays the result.

Some key features of this project include simple user input, quick conversion, no need for object creation, efficient memory usage, and basic error handling for invalid currency types. Overall, this project demonstrates how OOP concepts can be applied to solve real-world problems in an efficient and structured way.

Scope:

This project is suitable for basic currency conversion tasks and can be used by students, travellers, and individuals who need quick and simple currency calculations. It provides an easy way to convert major international currencies into Indian Rupees using predefined exchange rates.

The current system is simple and works without using any database or real-time data. However, it can be extended in the future by adding advanced features such as real-time exchange rate updates using APIs, support for more international currencies, and a graphical user interface (GUI) for better user experience.

System Requirements:

Hardware Requirements

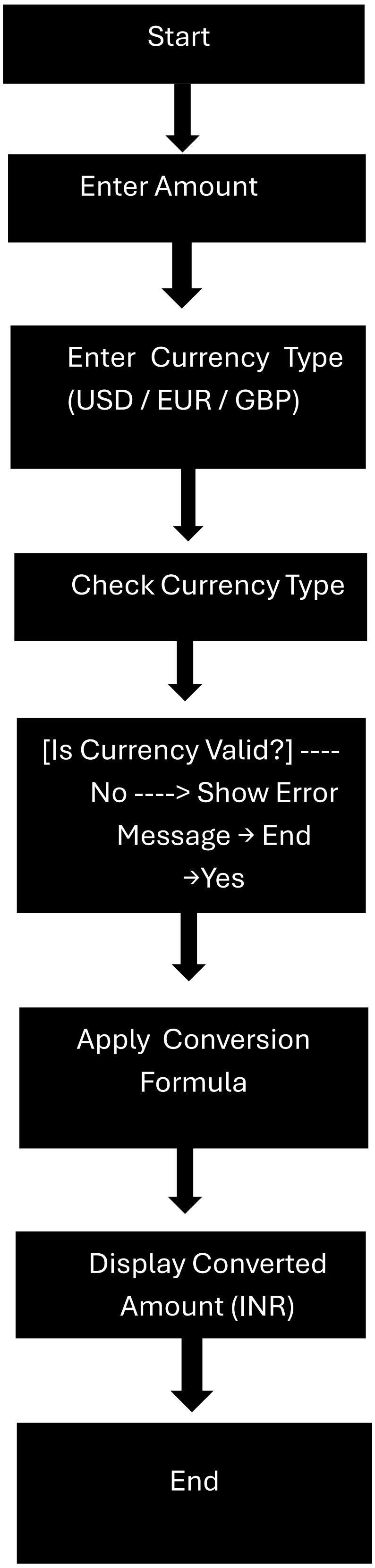
- Computer or Laptop
- Minimum 4 GB RAM
- Processor: i3 or above
- Hard Disk: At least 500 MB free space
- Keyboard and Monitor

Software Requirements

- Java Development Kit (JDK) for compiling and running the program
- Any Java IDE such as VS Code, Eclipse, or Edit Plus for writing and executing code
- Operating System: Windows, Linux, or macOS

Methodology :

- The Global Currency Converter project works in a simple step-by-step process using Java and OOP concepts. First, the program starts by defining a utility class that contains all the exchange rates as static constant variables. These rates are fixed and used throughout the program.
- Next, a static method is created to handle the conversion process. This method takes two inputs from the user: the amount of money and the type of currency (such as USD, EUR, or GBP). The program then checks the entered currency using conditional statements like if-else or switch.
- After identifying the correct currency, the program applies the corresponding exchange rate and multiplies it with the given amount to calculate the value in Indian Rupees (INR). If the user enters an invalid currency type, the program displays an error message.
- Finally, the converted result is displayed on the screen. The entire process is done without creating any object, as all operations are performed using static members. This makes the system simple, fast, and efficient.


Start
Enter Amount
Enter Currency Type (USD / EUR / GBP)
Check Currency Type
[Is Currency Valid?] ---- No ----> Show Error Message → End →Yes
Apply Conversion Formula
Display Converted Amount (INR)
End

Variables inside class

- double amount
- String currency

Methods inside class

- public static void main(String[] args)
- Takes input from user
- Calls Converter Engine To convert()
- Displays output

Array

- No array is used in this class

Implementation:

The Converter Engine class is a utility class that contains all the logic required for currency conversion. It stores exchange rates as static constant variables, which means the values remain fixed and can be accessed without creating any object. This class also contains a static method that takes the amount and currency type as input and performs the conversion. It checks the currency type using conditional statements and applies the correct exchange rate to return the converted value in INR. If an invalid currency is entered, it handles the error by displaying a message.

The **Main** class is responsible for interacting with the user. It takes input such as the amount and currency type from the user using the scanner. Then, it calls the conversion method from the Converter Engine class and displays the result on the screen.

Sample Input/Output

Input:

Enter Amount: 100

Enter Currency: USD

Output:

Converted Amount in INR: 8300.0

Future Enhancements:

This project can be improved in several ways. Real-time exchange rates can be added using APIs instead of fixed values. More currencies can be included to make the system more useful. A graphical user interface (GUI) can be developed to improve user experience. The system can also be converted into a web or mobile application for wider accessibility. Additionally, database integration can be used to store and update exchange rates automatically.

Conclusion:

The Global Currency Converter project successfully demonstrates the use of Object-Oriented Programming (OOP) concepts in Java, especially the use of static variables and static methods. The project provides a simple and functional system for converting different international currencies into Indian Rupees (INR) using predefined exchange rates.

Website :-

Oracle, "Java Documentation" . [Online]. Link : Java Software | Oracle